
ML-KEM.Encaps_internal(ek, m)

Uses the encapsulation key and randomness to generate a key and an associated ciphertext.

Input: encapsulation key $ek \in \mathbb{B}^{384k+32}$.

Input: randomness $m \in \mathbb{B}^{32}$.

Output: shared secret key $K \in \mathbb{B}^{32}$.

Output: ciphertext $c \in \mathbb{B}^{32(d_u k + d_v)}$.

1: $(K, r) \leftarrow G(m \| H(ek))$

2: $c \leftarrow \text{K-PKE.Encrypt}(ek, m, r)$

3: **return** (K, c)

▷ derive shared secret key K and randomness r

▷ encrypt m using K-PKE with randomness r
